ONCOLOGY · NEET-SS / FET / INI-SS

Soft-tissue sarcoma

1. Epidemiology and general features

- Soft-tissue sarcomas (STS) are malignant tumours of **mesenchymal origin** (fat, muscle, fibrous tissue, vessels, nerve sheath); **about 1% of adult cancers** but over **70 histological subtypes** (WHO 2020).
- **Benign soft-tissue tumours outnumber sarcomas by about 100:1**; **lipoma** is the commonest soft-tissue tumour.
- **Age:** any; most in adults over 50. In **children the commonest STS is rhabdomyosarcoma**.
- **Commonest STS in adults: undifferentiated pleomorphic sarcoma (UPS)** and **liposarcoma** are the two commonest (which is first varies by series; for extremity sarcomas in older adults UPS is often quoted, for all sites liposarcoma).
- **Site: extremities about 50-60%** (**lower limb, especially the thigh, is the commonest site**), trunk wall, **retroperitoneum about 15%**, head and neck, viscera.
- **Behaviour:** grow centrifugally and compress surrounding tissue into a **pseudocapsule (reactive zone)** containing **satellite tumour cells**, with **skip lesions** in high-grade tumours. **Shelling out through the pseudocapsule leaves tumour behind**, which is why wide excision is needed.
- **Spread: haematogenous**, **lungs commonest (about 80% of metastases)**. Lymph-node metastasis is **uncommon (under 5%) except in specific subtypes** (see below).
- **Predisposing factors:**
  - **Li-Fraumeni syndrome (TP53)**, **neurofibromatosis type 1** (MPNST), **hereditary retinoblastoma (RB1)**, **familial adenomatous polyposis/Gardner syndrome** (desmoid tumours), Werner syndrome.
  - **Prior radiotherapy** (radiation-associated sarcoma, latency usually over 5-10 years; angiosarcoma, UPS).
  - **Chronic lymphoedema** (**Stewart-Treves syndrome**: lymphangiosarcoma/angiosarcoma after mastectomy and axillary

clearance).
  - Chemicals (vinyl chloride: hepatic angiosarcoma), HIV/HHV-8 (Kaposi sarcoma).

2. Clinical evaluation of a soft-tissue mass

- Typical sarcoma: **painless, enlarging mass**; pain occurs in about a third. Patients often attribute it to trauma ("haematoma" that does not resolve).
- **Red-flag features of a soft-tissue lump (suspect sarcoma):**
  - **Size over 5 cm** (golf ball)
  - **Deep to the deep fascia**
  - **Increasing in size**
  - **Painful**
  - **Recurrence after previous excision**
- **Any one** feature warrants referral and imaging; deep or over 5 cm lumps have a high risk (about 10% or more of deep lumps over 5 cm are malignant). Superficial small lumps can still be sarcoma (e.g. epithelioid, myxofibrosarcoma).
- **Examination:** size, depth, consistency, mobility, transillumination, pulsatility, neurovascular status (Tinel sign over a nerve sheath tumour), **regional lymph nodes**, skin changes.

3. Investigations

- **Ultrasound:** first-line triage of superficial lumps (UK pathway); sarcoma suspected if **heterogeneous, vascular, deep, or over 5 cm**.
- **MRI with gadolinium: investigation of choice** for local evaluation: size, compartment, depth, relation to **neurovascular bundle and bone**, peritumoural oedema, necrosis; plans biopsy and surgery.
  - **Lipoma** is homogeneous fat signal, suppresses completely on fat-saturated sequences, with only thin septa.
  - **Atypical lipomatous tumour (ALT)**: thick (over 2 mm) or nodular septa, non-fatty areas, large size, deep location.
  - **Myxoid tumours** (myxoid liposarcoma, myxofibrosarcoma, myxoma) can look **cyst-like** (high T2) but **enhance** with contrast: never call a deep "cyst" benign without contrast.

- **Plain radiographs:** bone involvement, **calcification** (synovial sarcoma about 30%, phleboliths in haemangioma, myositis ossificans zonation), extraskeletal osteosarcoma.
- **Staging:**
  - **CT chest** for all intermediate- or high-grade sarcomas (lungs are the commonest metastatic site).
  - **Myxoid (round cell) liposarcoma:** metastasises to **extrapulmonary sites** (spine, retroperitoneum, other soft tissue) and may be missed by CT and PET: add **whole-spine MRI or whole-body MRI**.
  - **Lymph-node assessment** (clinical, ultrasound, CT/PET, sentinel node biopsy in selected cases) for node-seeking subtypes.
  - **CT abdomen/pelvis** for myxoid liposarcoma, retroperitoneal tumours and leiomyosarcoma; brain MRI for **alveolar soft part sarcoma** and angiosarcoma.
  - **PET-CT:** useful for high-grade tumours, rhabdomyosarcoma, NF1 (suspected malignant change in a plexiform neurofibroma) and response assessment.

Biopsy principles (Mankin)

- **Biopsy after all imaging**, at (or after discussion with) **the specialist sarcoma centre that will perform definitive surgery**. **Mankin et al. (1982, repeated 1996)**: biopsies performed in referring hospitals had **significantly more diagnostic errors, complications and changes in treatment (including unnecessary amputation)** than those performed in treating centres.
- **Image-guided core needle biopsy** is the standard (diagnostic accuracy over 90%); open incisional biopsy if core is non-diagnostic. **Fine-needle aspiration is not adequate** for primary diagnosis (no architecture or grade).
- **Rules:**
  - **Longitudinal** incision in the limb (in line with the definitive resection), as small as possible.
  - Through a **single compartment**; avoid intermuscular planes, joints and the **neurovascular bundle**.

- Meticulous haemostasis; if a drain is used, it exits **in line with and close to the incision**.
- Biopsy tract is considered contaminated and **excised en bloc** with the tumour.
- Sample the **enhancing, viable periphery** (avoid necrotic centre); send fresh tissue for **molecular genetics** and cultures where infection is possible.

- **Excisional biopsy** is acceptable only for **small (under 3-5 cm), superficial** lesions where re-excision would be straightforward.

4. Grading and staging

FNCLCC (French) grading system: the standard histological grade

Parameter	Score
Tumour differentiation	1 = closely resembles normal adult mesenchymal tissue; 2 = histological type certain; 3 = embryonal, undifferentiated, or type uncertain
Mitotic count (per 10 high-power fields)	1 = 0-9; 2 = 10-19; 3 = 20 or more
Tumour necrosis	0 = none; 1 = under 50%; 2 = 50% or more
Total score	**2-3 = Grade 1**; **4-5 = Grade 2**; **6-8 = Grade 3**

- **Grade is the most important predictor of distant metastasis** (and overall survival); **margin status is the most important predictor of local recurrence**.

AJCC 8th edition: trunk and extremity sarcoma

T stage	Size
T1	5 cm or less
T2	Over 5 cm up to 10 cm
T3	Over 10 cm up to 15 cm
T4	Over 15 cm

Stage	Definition
IA	T1, N0, M0, grade 1 (or GX)
IB	T2-T4, N0, M0, grade 1 (or GX)
II	T1, N0, M0, grade 2-3
IIIA	T2, N0, M0, grade 2-3
IIIB	T3-T4, N0, M0, grade 2-3
IV	**Any N1 or any M1**

- **Depth (superficial vs deep) was removed from the AJCC 8th edition staging** (it was in the 7th), though it remains prognostic and is still a red flag.
- **Retroperitoneal sarcomas** have a separate AJCC staging; Enneking staging (low/high grade, intra/extracompartmental, metastasis) is still used by orthopaedic oncologists, particularly for musculoskeletal surgical planning.

5. Important subtypes (high-yield table)

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Atypical lipomatous tumour / well-differentiated liposarcoma	Older adults; **ALT in limbs**, WDL in retroperitoneum	**MDM2 and CDK4 amplification (12q13-15 ring/giant marker chromosomes)**; MDM2 FISH distinguishes from lipoma	**Does not metastasise**; local recurrence; **marginal excision acceptable in limbs**; risk of **dedifferentiation** (higher in retroperitoneum)

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Myxoid liposarcoma (including round-cell)	Young-middle-aged adults (30-50), **thigh**	**t(12;16) FUS-DDIT3** (rarely t(12;22) EWSR1-DDIT3); **"chicken-wire" capillaries**, lipoblasts	**Very radiosensitive** and chemosensitive (trabectedin); **extrapulmonary metastases** (spine, retroperitoneum, soft tissues); round-cell over 5% = higher grade
Pleomorphic and dedifferentiated liposarcoma	Older adults	Pleomorphic lipoblasts; dedifferentiated keeps MDM2 amplification	High grade
Undifferentiated pleomorphic sarcoma (UPS)	**Elderly (over 50-60), thigh**, deep	Diagnosis of exclusion; formerly "malignant fibrous histiocytoma"	High grade; lung metastases; also arises in bone (infarct, Paget, radiation)
Myxofibrosarcoma	Elderly, **superficial**, limbs	Myxoid stroma, curvilinear vessels	**Infiltrative "tail sign" along fascia on MRI**; **very high local recurrence**; needs wide margins

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Synovial sarcoma	**Adolescents and young adults (15-40)**, **around the knee (popliteal fossa)**, foot and ankle; **rarely intra-articular** despite the name	**t(X;18)(p11;q11) SS18-SSX1/SSX2**; **TLE1** and **SS18-SSX fusion-specific antibody** positive; biphasic or monophasic	Often **slow-growing and painful**, may be small and mistaken for benign; **calcification in about 30%** on radiographs; **chemosensitive (ifosfamide)**; lymph-node metastases; late metastases
Epithelioid sarcoma	**Young adults, hand, wrist and forearm (commonest STS of the hand in many series)**	**Loss of INI1 (SMARCB1)**; keratin and EMA positive	Ulcerating nodules mistaken for **granuloma, infection or rheumatoid nodule**; spreads along fascia and tendons; **lymph-node metastases**; tazemetostat (EZH2 inhibitor)
Clear cell sarcoma ("melanoma of soft parts")	Young adults, **foot and ankle tendons and aponeuroses**	**t(12;22) EWSR1-ATF1**; **S100, HMB-45, Melan-A positive** (melanocytic markers)	Lymph-node metastases; poor prognosis; distinguished from melanoma by the fusion

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Alveolar soft part sarcoma	Adolescents and young adults, thigh/buttock	**t(X;17) ASPSCR1-TFE3**; TFE3 nuclear	Highly vascular; **metastases to lung and brain** early; responds to immune checkpoint inhibitors
Malignant peripheral nerve sheath tumour (MPNST)	About half in **NF1** (lifetime risk about 8-13%), sciatic nerve, brachial plexus	S100 patchy or lost, loss of H3K27me3	Suspect when a **plexiform neurofibroma becomes painful or grows rapidly, or a new neurological deficit appears**; FDG-PET (high SUV) guides biopsy; poor prognosis
Leiomyosarcoma	Middle-aged and elderly; **retroperitoneum, IVC**, uterus, limbs (vascular origin)	Desmin, SMA, h-caldesmon	Haematogenous spread
Angiosarcoma	Elderly scalp, post-radiation breast, **Stewart-Treves** (lymphoedematous arm)	CD31, ERG, CD34	Very aggressive, multifocal
Dermatofibrosarcoma protuberans (DFSP)	Young-middle-aged adults, trunk, superficial	**COL1A1-PDGFB**, CD34 positive	Locally aggressive, rarely metastasises; wide excision or **Mohs**; **imatinib** for unresectable

Subtype	Typical patient and site	Key pathology/genetics	Behaviour and exam points
Rhabdomyosarcoma	Children; alveolar type in adolescent extremities	Desmin, myogenin; PAX3/7-FOXO1	VAC chemotherapy; lymph nodes
Extraskeletal myxoid chondrosarcoma	Adults, thigh	NR4A3 rearrangement (EWSR1-NR4A3)	Slow but metastasises

Lymph-node-seeking sarcomas (classic list, mnemonic "SCARE")

- **S**ynovial sarcoma, **C**lear cell sarcoma, **A**ngiosarcoma, **R**habdomyosarcoma, **E**pithelioid sarcoma (alveolar soft part sarcoma also).

> Exam pearl: Lymph node spread is rare in sarcoma except SCARE (synovial, clear cell, angiosarcoma, rhabdomyosarcoma, epithelioid). Myxoid liposarcoma spreads to extrapulmonary sites, so stage with whole-spine or whole-body MRI. Grade predicts metastasis; margin predicts local recurrence.

6. Intermediate (locally aggressive) tumours: desmoid fibromatosis

- **Desmoid-type (aggressive) fibromatosis:** clonal myofibroblastic proliferation; **locally infiltrative, never metastasises**, but recurs often and can cause morbidity.
- **Genetics:** sporadic tumours have **CTNNB1 (beta-catenin) mutations** (S45F mutation associated with higher recurrence); **APC germline mutation** in **familial adenomatous polyposis/Gardner syndrome** (intra-abdominal, mesenteric desmoids). **Nuclear beta-catenin** on immunohistochemistry.
- Sites: shoulder girdle, chest wall, thigh, abdominal wall (often **postpartum**; oestrogen influence), mesentery.
- **Natural history is unpredictable**: a substantial proportion stabilise or **regress spontaneously**.
- **Management (Desmoid Tumor Working Group global consensus 2020):**
  - **Active surveillance first-line** for most asymptomatic or non-threatening tumours, with MRI after 1-2 months and then every 3-6 months.

- Treat only if **persistent progression** or symptoms, or threat to vital structures.
- **Systemic therapy is preferred over surgery** for progressive disease: **nirogacestat** (gamma-secretase inhibitor; **DeFi trial**, NEJM 2023, improved progression-free survival), **sorafenib** (Gounder et al., NEJM 2018), low-dose methotrexate-vinblastine, pegylated liposomal doxorubicin, NSAIDs/hormonal agents (weak evidence).
- **Surgery** (aiming for negative margins without major functional loss) for selected progressive limb or abdominal-wall tumours; **positive margins are often watched** rather than re-excised. Radiotherapy, cryoablation are alternatives.

7. Management of soft-tissue sarcoma

Surgery: wide local excision is the mainstay

- **Margins (Enneking):**

Margin	Plane	Local recurrence (without radiotherapy)
Intralesional	Through the tumour	Almost certain
Marginal	Through the **reactive zone/pseudocapsule**	High for high-grade tumours (satellite cells)
Wide	Through normal tissue with a **cuff of normal tissue** around the reactive zone	Acceptable, especially combined with radiotherapy
Radical	**Entire compartment** removed (or amputation proximal to the compartment)	Lowest; rarely needed now

- In practice, a **wide margin of about 1-2 cm of normal tissue, or an intact fascial barrier** (fascia, periosteum, epineurium, vessel adventitia act as barriers), is aimed for. **R0** = microscopically negative, **R1** = microscopic tumour at margin, **R2** = macroscopic tumour left.
- **Limb salvage is possible in over 90% of extremity sarcomas**. **Rosenberg et al. (NCI, 1982): limb-sparing surgery plus radiotherapy gave the same**

overall survival as amputation (higher local recurrence, but no survival difference). This established limb salvage.

- **Amputation** indications: tumour encasing major neurovascular structures where reconstruction is impossible or would leave a useless limb, extensive contamination (failed previous surgery, fungating tumour), recurrence after maximum radiotherapy, patient choice.
- Neurovascular involvement: **vascular resection and reconstruction** (graft) is possible; **epineurial dissection** if the nerve is abutted, nerve resection if encased (sciatic resection may still give a functional limb).
- Soft-tissue reconstruction: local or **free flaps** (especially after preoperative radiotherapy to reduce wound complications).

Unplanned excision ("whoops procedure")

- Excision of a mass without preoperative imaging and assumed to be benign, later found to be a sarcoma; usually **intralesional or marginal**.
- **Residual tumour is found in about 40-50% of re-excision specimens**.
- Management: refer to a sarcoma centre, **MRI (and staging)**, then **re-excision of the entire tumour bed and scar with wide margins**, usually with radiotherapy. Prognosis after appropriate re-excision approaches that of planned surgery, but morbidity (larger resections, flaps) is higher.

Radiotherapy

- **Indications:** most **intermediate- and high-grade, deep or over 5 cm** tumours, and **close or positive margins**. Low-grade, small, superficial tumours excised with wide margins may need surgery alone.
- **Yang et al. (NCI, 1998)** randomised trial: adjuvant external-beam radiotherapy after limb-sparing surgery **reduced local recurrence but did not change overall survival**.
- **Preoperative versus postoperative radiotherapy: O'Sullivan et al. (NCIC SR2, Lancet 2002):**

Feature	Preoperative RT (50 Gy)	Postoperative RT (60-66 Gy)
Major wound complications	**Higher (about 35%)**	Lower (about 17%)

Feature	Preoperative RT (50 Gy)	Postoperative RT (60-66 Gy)
Late toxicity (fibrosis, joint stiffness, oedema)	**Lower** (Davis 2005)	**Higher**
Dose and field	Lower dose, smaller field	Higher dose, larger field (whole surgical bed)
Local control and survival	**Equivalent**	Equivalent
Other	Tumour may shrink or become better defined; thigh wounds worst	Full pathology available first

    - **Current practice favours preoperative radiotherapy** for most extremity STS, because **late toxicity is permanent whereas wound complications are temporary and treatable** (plan plastic surgical closure). Wound complications are commonest in the **lower limb, especially the thigh**.
    - Preoperative radiotherapy is particularly useful for **myxoid liposarcoma** (very radiosensitive, marked shrinkage).
- **Retroperitoneal sarcoma:** the **STRASS trial (EORTC 62092)** showed **no overall benefit of preoperative radiotherapy** on abdominal recurrence-free survival (a possible benefit in liposarcoma subgroups is debated); surgery is the main treatment (en bloc resection with adjacent organs where needed).
- **Brachytherapy** and **intensity-modulated radiotherapy (IMRT)** reduce dose to normal tissue.

Chemotherapy

- **Standard drugs: doxorubicin with or without ifosfamide**. Newer: trabectedin (myxoid/round cell liposarcoma, leiomyosarcoma), eribulin (liposarcoma), gemcitabine-docetaxel, pazopanib (non-adipocytic sarcomas), **imatinib (GIST, DFSP)**, immune checkpoint inhibitors (UPS, alveolar soft part sarcoma).
- **Adjuvant chemotherapy for adult STS is controversial**: the **EORTC 62931** trial showed **no benefit** of adjuvant doxorubicin-ifosfamide; meta-analyses (e.g. Sarcoma Meta-analysis Collaboration, Pervaiz 2008) show a **small**

benefit in local and distant recurrence with a marginal overall survival benefit. Generally offered for **high-risk (large, high-grade, deep) tumours of chemosensitive subtypes**, often as neoadjuvant therapy (ISG-STS 1001 supports histology-tailored or standard anthracycline-ifosfamide neoadjuvant regimens).

- **Chemotherapy is a standard component** for **rhabdomyosarcoma, extraskeletal Ewing sarcoma** and is commonly used in **synovial sarcoma** and **myxoid liposarcoma** (chemosensitive).
- **Isolated limb perfusion** (hyperthermic, **melphalan plus TNF-alpha**) for locally advanced, otherwise unresectable extremity sarcomas to avoid amputation.

Metastatic disease

- **Pulmonary metastasectomy** for limited, resectable lung metastases with a long disease-free interval (about 20-30% long-term survival in selected patients).
- Palliative chemotherapy, radiotherapy, targeted agents.

Follow-up

- Local recurrence: clinical examination and **MRI** (or ultrasound) of the primary site; distant: **CT chest** (or chest radiograph); every 3-4 months for 2-3 years for high-grade tumours, then 6-monthly to 5 years, then annually (most recurrences occur within 2-3 years; some subtypes recur late, e.g. synovial sarcoma, ALT).

8. Prognosis

- **Overall 5-year survival about 60-65%** for extremity STS (higher for low-grade, under 50% for large high-grade tumours).
- **Predictors of distant metastasis and death: high grade**, **large size (over 5 cm, especially over 10 cm)**, **deep location**, older age, certain subtypes (MPNST, angiosarcoma, pleomorphic liposarcoma, UPS), **local recurrence** (marker of aggressive biology).
- **Predictors of local recurrence: positive margins (the most important)**, high grade, no radiotherapy, myxofibrosarcoma/infiltrative types, previous recurrence.
- Nomograms (e.g. **Sarculator**, MSKCC) estimate individual risk.

9. Differential diagnosis of soft-tissue masses (distractors)

Lesion	Clues
Lipoma	Superficial, soft, homogeneous fat on MRI, suppresses on fat-saturation; intramuscular lipoma infiltrates muscle but is still benign
Haematoma	History of trauma or anticoagulation, should resolve; **a "haematoma" that does not resolve must be imaged with contrast MRI** (haemorrhagic sarcoma)
Myositis ossificans	After trauma; **zonal pattern: mature peripheral ossification with an immature centre** (opposite of osteosarcoma); early biopsy may mislead (pseudosarcomatous); calcification appears at 3-4 weeks
Nodular fasciitis	Rapidly growing (weeks) superficial nodule in young adults, forearm; **USP6 rearrangement**; self-limiting
Schwannoma / neurofibroma	Eccentric (schwannoma) or fusiform central (neurofibroma) along a nerve; **"split-fat", "target" and "fascicular" signs** on MRI; positive Tinel
Haemangioma / vascular malformation	Phleboliths, varies with position, serpiginous flow voids
Intramuscular myxoma	Homogeneous, very high T2, thin peripheral fat rim; with fibrous dysplasia in **Mazabraud syndrome**
Ganglion, Baker cyst	Thin-walled, peripheral enhancement only, communication with joint
Tenosynovial giant cell tumour (PVNS)	Low signal on T2 and **blooming** on gradient echo (haemosiderin); CSF1 overexpression; pexidartinib
Abscess, tuberculous cold abscess	Systemic features, rim enhancement, fluid on aspiration

Exam pearl: A soft-tissue lump that is over 5 cm, deep to fascia, enlarging, painful or recurrent needs contrast MRI and then a core needle biopsy at the sarcoma centre that will do the surgery, never an excision. Treatment is wide excision with radiotherapy (preoperative preferred: more wound complications but less permanent late toxicity, per O'Sullivan NCIC SR2); limb salvage equals amputation for survival (Rosenberg 1982).

Written by Claude for Ortho preSSaleel · updated 2026-11-15. Check doubtful points in a standard textbook.